Akoizo Original Practice - Form 4 - Answer key

Division B | Ping Pong Parachute | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) An observer 8 m from the launcher sees apogee at 45°, with eye height 1.5 m. What is the apogee?

A. 11.3 m

B. 6.5 m

C. 9.5 m

D. 8 m

Correct: C

2. (2 points) Raising launch pressure mainly increases:

Unit: How an air-powered bottle rocket climbs.
Worked-case data: A rocket reaches apogee 1.2 s after launch, and the payload then descends for 11.5 s. What share of time aloft came from the descent?
Separate investigation observations:
40 psi: Peak about 12 ft; time aloft 8.1 s.
55 psi: Peak about 18 ft; time aloft 12.4 s.
70 psi: Peak about 24 ft; time aloft 16.0 s in a 30 ft hall.

A. The parachute area

B. The nozzle diameter

C. The impulse and launch speed

D. The rocket’s mass

Correct: C

3. (2 points) Why is changing one variable at a time important in test flights?

A. It shows which change caused the difference in results

B. The rules require it

C. It makes rockets lighter

D. It shortens the log

Correct: A

4. (2 points) Why do many canopies include a small hole at the top?

Unit: Parachute physics: drag and slow descent.
Worked-case data: A 4.7 g payload uses a 40 cm diameter dome (C_d about 1.5). Estimate its descent speed and the time to fall 7 m.
Separate investigation observations:
30 cm canopy: Opens every time; 7.8 s descent.
60 cm canopy: Opens 6 of 10 times; 14.9 s when open, 3 s when not.
50 cm canopy, looser fold: Opens 10 of 10; 12.6 s descent.

A. To reduce swinging and steady the descent

B. To make the parachute fall faster for scoring

C. Because the rules require it

D. To attach the ping pong ball

Correct: A

5. (4 points) Why are cutting, sanding, and gluing the bottle prohibited?

Unit: Building a legal, stable rocket.
Worked-case data: A rocket's CG is 18 cm from the nose and its CP is 21 cm from the nose. The bottle diameter is 8 cm. Is it stable enough?
Separate investigation observations:
Flight 1: Rocket corkscrews and lands near a wall.
Inspection: One fin is twisted about 8° from the body line.
Flight 2: Fin realigned: rocket flies straight up.

answer: They weaken the pressure vessel, which could burst under pressure and injure people. (4 points).

6. (4 points) Explain why hitting the ceiling is usually worse than flying lower.

answer: Timing stops at contact and the launch is multiplied by 0.25, so even a high flight scores only a fraction of a modest clean one. (4 points).

7. (4 points) Describe how a passive nose cup releases the payload.

Unit: Deployment and separation.
Worked-case data: Video shows a payload leaving the rocket 0.3 s after launch, well before apogee, and the parachute tangling. What should change?
Separate investigation observations:
Cup depth 2 cm: Payload left during climb in 4 of 5 flights.
Cup depth 4 cm: Payload stayed attached in 3 of 5 flights.
Cup depth 3 cm: Payload released near apogee in 5 of 5 flights.

answer: The payload rides in the cup during thrust; as the rocket slows and tips over at apogee, the payload's momentum and drag carry it out of the open cup. (4 points).

8. (4 points) Why record peak height in a gym that is lower than the competition venue?

Unit: Apogee, ceiling height, and choosing pressure.
Worked-case data: Your chart shows 50 psi -> 17 ft, 60 psi -> 21 ft, 70 psi -> 24 ft, with about ±1.5 ft spread. The ceiling is 25 ft. Which pressure should you choose?
Separate investigation observations:
Chart: 45 psi -> 15 ft, 55 psi -> 19 ft, 65 psi -> 22 ft.
Spread: About ±1 ft at each pressure.
Venue: Ceiling announced at 21 ft.

answer: Your pressure-height data lets you pick a safe pressure for whatever ceiling is announced, even if it differs from your practice room. (4 points).

9. (4 points) Why is maximum height not the goal?

Unit: How an air-powered bottle rocket climbs.
Worked-case data: A rocket reaches apogee 1.2 s after launch, and the payload then descends for 11.5 s. What share of time aloft came from the descent?
Separate investigation observations:
40 psi: Peak about 12 ft; time aloft 8.1 s.
55 psi: Peak about 18 ft; time aloft 12.4 s.
70 psi: Peak about 24 ft; time aloft 16.0 s in a 30 ft hall.

answer: Flying too high risks the ceiling penalty; the goal is a safe apogee below the ceiling followed by the slowest possible descent. (4 points).

10. (4 points) What should a team do on launch 2 after a strong launch 1?

answer: Repeat the same proven setup to secure a strong tiebreaker time rather than take a big risk. (4 points).

11. (6 points) Solve this case and explain your reasoning: A 4.7 g payload uses a 40 cm diameter dome (C_d about 1.5). Estimate its descent speed and the time to fall 7 m.

reasoning: W = 0.0047 x 9.8 about 0.046 N; A = π x 0.20^2 about 0.126 m^2.
v = sqrt (2 x 0.046 ÷ (1.2 x 1.5 x 0.126)) about 0.64 m/s.
Time about 7 ÷ 0.64 about 11 s (plus the climb). (3 points).

conclusion: A 40 cm canopy turns a 7 m drop into about 11 seconds; doubling diameter would nearly double that. (3 points).

12. (2 points) Define “Pressure vessel” in the context of Ping-Pong Parachute.

answer: The unaltered bottle that holds the compressed air. (2 points).

13. (2 points) Define “Time aloft” in the context of Ping-Pong Parachute.

answer: Time from leaving the launcher until the payload lands. (2 points).

14. (2 points) Define “Separation” in the context of Ping-Pong Parachute.

answer: Payload leaving the rocket during flight. (2 points).

15. (2 points) Define “Tangent method” in the context of Ping-Pong Parachute.

answer: Height = distance x tan(angle) + eye height. (2 points).

16. (2 points) Define “Stability margin” in the context of Ping-Pong Parachute.

answer: Distance the CG sits ahead of the CP, often in body diameters. (2 points).

17. (2 points) Define “Multiplier” in the context of Ping-Pong Parachute.

answer: A factor applied to a launch time as a penalty. (2 points).

18. (2 points) Define “Burnout” in the context of Ping-Pong Parachute.

answer: The moment the air thrust ends. (2 points).